

# Data Analyst Capstone Project Report

## Developer Technology Skills Analysis

From data collection and wrangling to IBM Cognos Analytics

### Learner

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### Date

September 26, 2026

### Dataset

Stack Overflow Developer Survey  
18,845 responses • 114 variables

# Outline

Presentation structure aligned to the capstone grading requirements

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# Executive Summary

Key findings from the developer technology survey

- JavaScript is the most frequently reported current language (14,943 respondents), followed by SQL and HTML/CSS.
- PostgreSQL leads current database usage (11,514 respondents); MySQL and SQLite are also prominent.
- Future preferences continue to include established technologies, while Go and Rust appear in the desired-language top 10.
- The Cognos dashboard combines technology usage, future preferences, and demographic patterns across three tabs.
- Survey preferences are useful for understanding the developer ecosystem but should be combined with job-posting and training-market data for broader demand analysis.

# Introduction

Purpose, target audience, and value of the analysis

## Purpose

Analyze current technology usage and future technology preferences among developers using the latest Stack Overflow Developer Survey dataset.

## Target Audience

Data analysts, technology leaders, workforce planners, educators, and teams monitoring developer skill trends.

## Business Value

Provide evidence for technology planning, skills development, training priorities, and further market research.

- Key questions: Which languages and databases are widely used? Which technologies are developers interested in for the future?
- The dashboard also examines platforms, web frameworks, age, country, education level, and age by education.

# Methodology

Data sources, collection methods, and key wrangling steps

## Data source

Stack Overflow  
Developer Survey  
**18,845**  
responses; 114  
variables

1

## Collection

Public dataset;  
project workflow  
also reviewed APIs  
and web scraping  
as data-collection  
methods

2

## Quality checks

Data types, missing  
values, duplicates,  
distributions, and  
outliers

3

## Wrangling

Numeric  
conversion,  
imputation/normali  
zation where  
appropriate, and  
splitting semicolon-  
separated  
technology fields

4

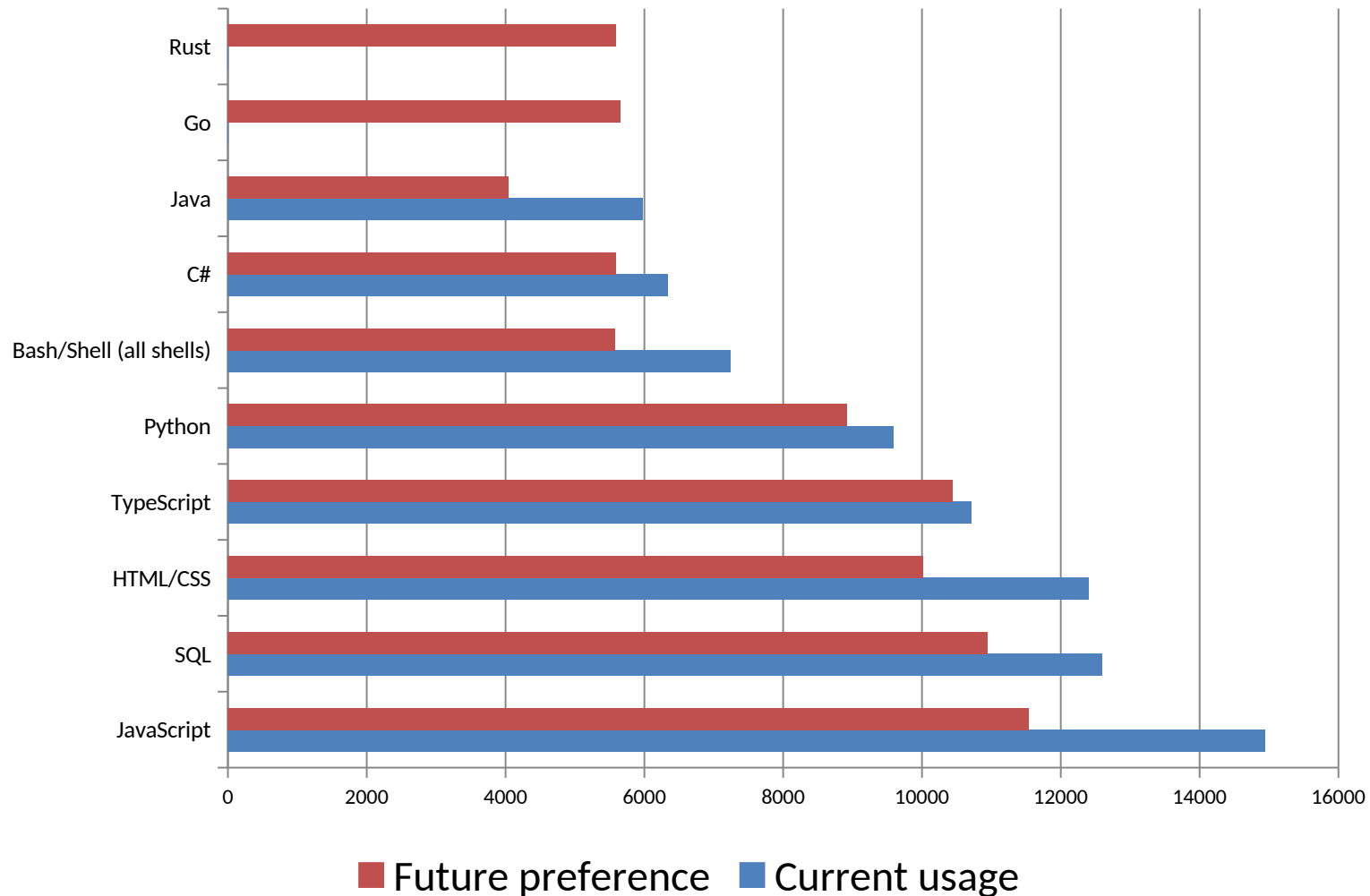
## Analysis & BI

Counts,  
distributions,  
comparisons,  
charts, and IBM  
Cognos Analytics  
dashboard

5

# Programming Languages Trends

Top 10 current usage versus next-year/future preference



## Current top 10

JavaScript 14,943  
SQL 12,602  
HTML/CSS 12,410  
TypeScript 10,709  
Python 9,590  
Bash/Shell 7,244  
C# 6,340  
Java 5,948

## Future top 10

JavaScript 11,541  
SQL 10,944  
TypeScript 10,437  
HTML/CSS 10,016  
Python 8,919  
Go 5,661  
Rust 5,597  
C# 5,590  
Bash/Shell 5,582  
Java 4,048

# Programming Languages Trends — Findings and Implications

What the current and future preference comparison indicates

## Finding 1

JavaScript, SQL, HTML/CSS, TypeScript, and Python remain prominent across both measures.

## Finding 2

The desired-language list includes Go and Rust in the top 10, indicating interest beyond established mainstream choices.

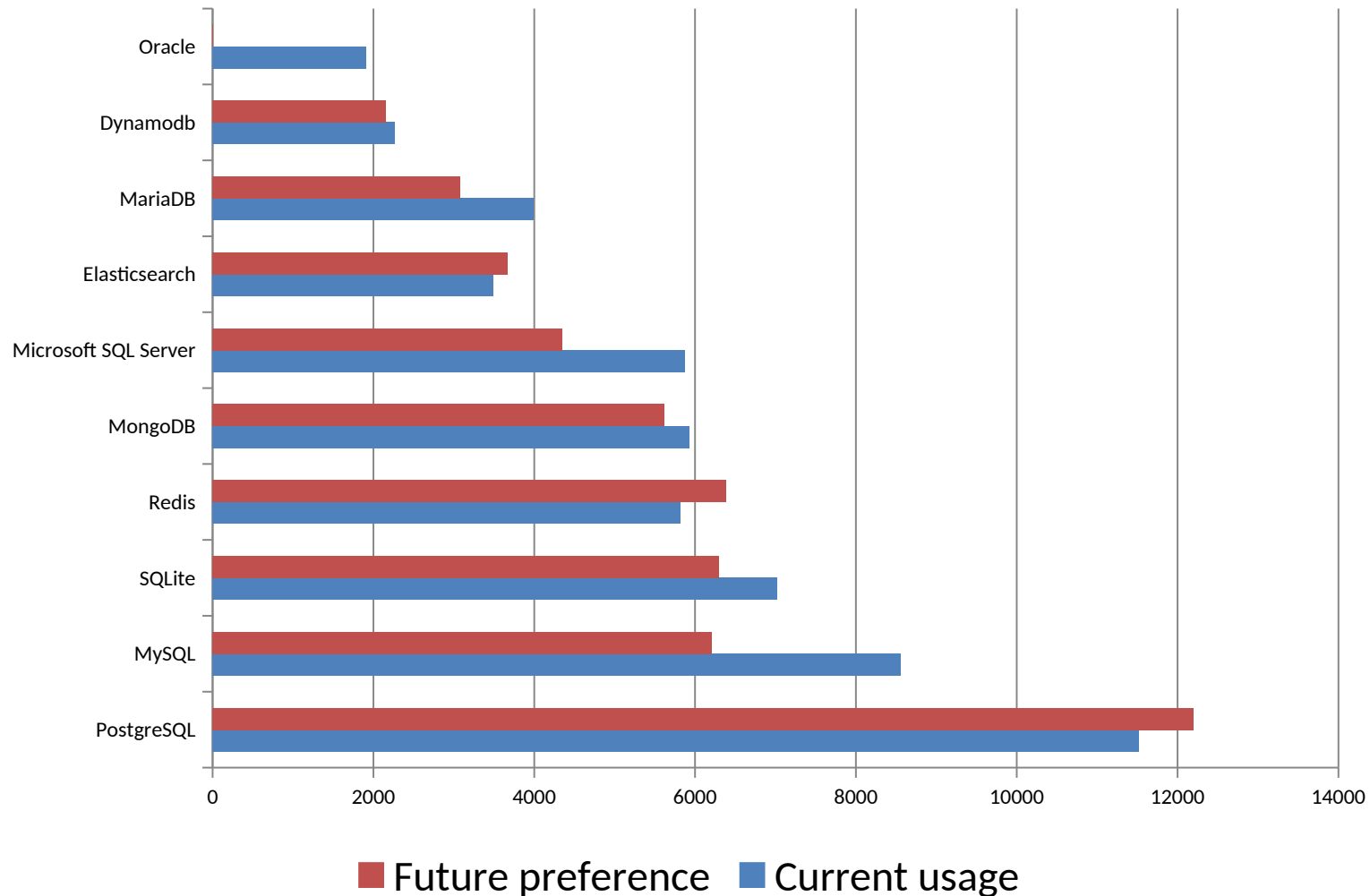
## Finding 3

Future preference counts are lower than current-use counts for several technologies because they represent a different survey question.

- Implication for skills planning: maintain strong foundations in widely used technologies while monitoring emerging preferences such as Go and Rust.
- Implication for training: use current usage to support existing-skill needs and future preference to identify learning/adoption interests.
- Limitation: these are developer-reported survey measures, not direct job-posting demand.

# Database Trends

Top 10 current databases versus future preference



## Current top 10

PostgreSQL 11,514  
MySQL 8,556  
SQLite 7,021  
MongoDB 5,930  
Microsoft SQL Server 5,870  
Redis 5,814  
MariaDB 3,994

## Future top 10

PostgreSQL 12,193  
Redis 6,384  
SQLite 6,295  
MySQL 6,204  
MongoDB 5,618  
Microsoft SQL Server 4,345  
Elasticsearch 3,665  
MariaDB 3,078  
Dynamodb 2,154  
Supabase 1,623

# Database Trends — Findings and Implications

Interpretation of current database usage and future preference

## Finding 1

PostgreSQL leads both current usage and future preference in this survey.

## Finding 2

Redis has a notably strong future-preference position compared with its current-use position.

## Finding 3

Supabase appears in the future top 10 while Oracle appears in the current top 10, showing some movement in preferences.

- Implication: PostgreSQL, SQL, and widely used relational/database skills remain relevant to the surveyed developer population.
- Implication: teams planning future learning or experimentation can monitor technologies with strong future preference, including Redis and Supabase.
- Market-demand caution: survey preference should be triangulated with job postings, training portals, and other market sources.

# IBM Cognos Analytics Dashboard

Three required tabs and the visualizations they contain

## TAB 1 — Current Technology Usage

Top 10 programming languages • Top 10 databases • Top 10 platforms • Top 10 web frameworks

Visuals: bar, column, word cloud, hierarchy bubble.

## TAB 2 — Future Technology Trend

Top 10 desired programming languages • Top 10 desired databases • Top 10 desired platforms • Top 10 desired web frameworks

Visuals: bar, column, tree map, hierarchy bubble.

## TAB 3 — Demographics

Age distribution • respondent count by country • formal education level • age classified by education level

Visuals: pie, map, line, stacked bar.

**Dashboard purpose: provide an interactive view of technology usage, future preferences, and respondent demographics.**

# Discussion

Key insights derived from the dashboard analysis

- Current technology usage is concentrated around widely established languages, databases, platforms, and frameworks.
- Future-preference views preserve several mainstream technologies while highlighting technologies developers want to use or learn.
- The three dashboard tabs connect technology trends with respondent demographics, helping users explore the survey population in context.
- The dashboard is descriptive: it identifies patterns in survey responses rather than proving causal relationships or forecasting employment demand.
- Combining these findings with external labor-market data would strengthen workforce and training decisions.

# Overall Findings and Implications

Connecting the results to broader technology and workforce planning

## Technology continuity

Core technologies such as JavaScript, SQL, HTML/CSS, Python, and PostgreSQL remain highly represented in the survey.

## Technology evolution

Future preferences include emerging or expanding interests such as Go, Rust, Redis, and Supabase.

## Skills planning

Organizations can use current-use data for baseline skills and future-preference data to identify areas worth monitoring.

## Evidence-based decisions

Survey evidence should be combined with job postings, training portals, and market data before making workforce decisions.

**Broader implication: technology skills planning benefits from continuously updated, multi-source evidence rather than a single survey snapshot.**

# Conclusion

## Project conclusions

- The project successfully analyzes current developer technology usage and future technology preferences using the Stack Overflow Developer Survey.
- Programming-language analysis shows strong representation of JavaScript, SQL, HTML/CSS, TypeScript, and Python, with Go and Rust among notable future preferences.
- Database analysis shows PostgreSQL as a leading technology in both current and future-preference measures, with other shifts in relative preference.
- IBM Cognos Analytics provides a three-tab dashboard covering current technology, future trends, and demographics.
- The results can support skills and training discussions, but broader industry decisions should incorporate additional market evidence.
- End-to-end workflow demonstrated: Collect → Clean → Explore → Analyze → Visualize → Communicate.

# Thank You

Data Analyst Capstone Project Report

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**Questions & Discussion**

IBM Data Analyst Professional Certificate